MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

AMPLIFIER, SUMMING, ANALOG, KEY CODE BED

This specification is approved for use by Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the procurement of an electronic summing amplifier module.

1.2 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Name	Amplifier, summing
Family	Analog
Key code	BED
Size	1A
Weight	35 grams maximum
Failure rate	0.290 failures/10^6 hours maximum
Environment	Class I as specified in MIL-STD-1389
Maximum power dissipation	693 mW
Supply voltages	±15 V ±10 percent
Maximum input signal	Supply voltage or ±15 V_p
Output impedance	15Ω maximum at 10 kHz
Gain	Programmable (unity, 1/2)
Gain accuracy	±0.5 dB maximum
Input offset voltage	6 mV maximum
Bandwidth	75 kHz minimum
Output slew rate	0.3 V/µs minimum

1.3 Module function. The summing amplifier module, designated as key code BED, is a Standard Electronic Modules Program (SEMP) electronic module with 0.100 inch (2.54 mm) grid center contacts. This module contains six independent inverting amplifiers. The two summing inputs on each amplifier give a unity gain or a gain of one half. The summing node on each amplifier is brought out for gain programming.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command, ATTN: ELEX 5043, Department of the Navy, Washington, D.C. 20360 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(All military standards and military specifications shall be ordered from the Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.)

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 0.290 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be BED.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

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3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/205-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier.)

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance test. All production modules shall be tested in accordance with notes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed two years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. The following test equipment is necessary to conduct acceptance tests for this module. Test equipment shall be as specified in MIL-STD-1665 and that listed below:

Equipment	MIL-STD-1665 designator	No. required (maximum per test figure
DC power supply	Item P01	3
Sine wave generator	Item G01	1
Megohm bridge	Item R06	1
Ohmmeter	Item R01	1
Digital multifunction meter	Item M02	1
AC voltmeter	Item W05	1
Counter timer	Item T01	1
DC voltmeter	Item V01	1
Phase angle meter	Item Z03	1
Distortion analyzer	Item B01	1
Capacitance bridge	Item C01	1
DC ammeter	Item X01	1
Spectrum analyzer	Item B03	1
Square wave generator	Item H01	1
Oscilloscope	Item D01	1
Oscilloscope probe	Item Y02	1
Oscilloscope probe	Item Y04	1
Switch		
SPST		1

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Resistors	No. required
2 kΩ ±1%, 1/8 W	6
4.75 kΩ ±5%, 1/8 W	1
20 kΩ ±0.1%, 1/8 W	1
47Ω ±5%, 1/4 W	2
100Ω ±0.1%, 1/8 W	1
Capacitor	
20 µF ±20%, 20 Vdc	2

All resistors shall be metal film.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the SEMP Quality Assurance Activity (SEMP-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through U shall be performed in any order with the following exceptions. If conditions A or B are part of a sequence of tests, the order of testing shall be as follows: A, B where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

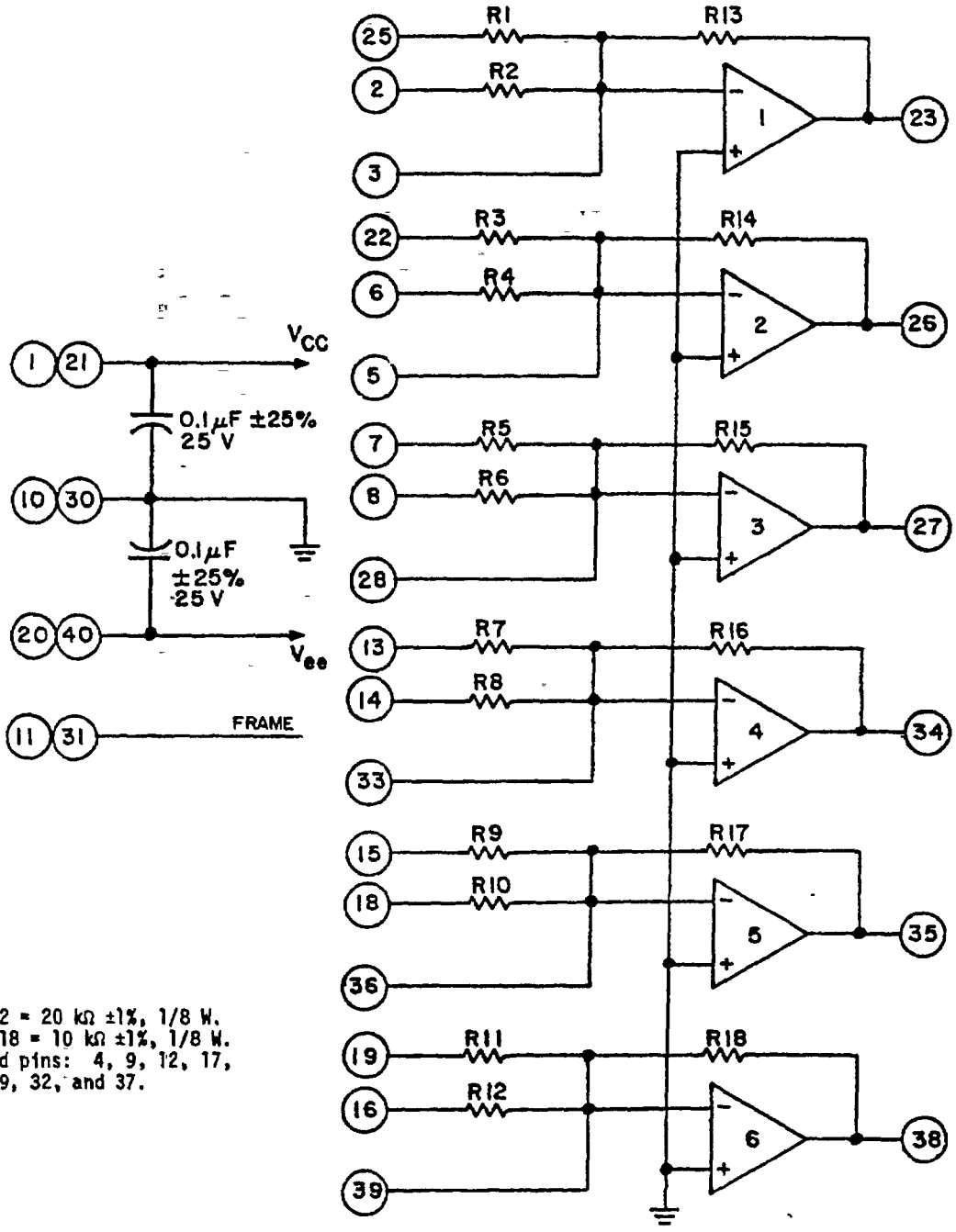
4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through U in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ± 5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions C through N and R through U shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 3. Following the vibration tests, the electrical tests of table I which are marked with an (X) in the test type column shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 4. Following the life test and while still at maximum class temperature, test conditions C through N and R through U shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of module temperature to 25°C, test conditions A and C through U shall be performed in the order specified in 4.4.2 using the 25°C ambient end-of-life requirements of table I.

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NOTES:
1. R1-R12 = 20 kΩ ±1%, 1/8 W.
2. R13-R18 = 10 kΩ ±1%, 1/8 W.
3. Unused pins: 4, 9, 12, 17, 24, 29, 32, and 37.

FIGURE 1. BED module function diagram, amplifier, summing.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° ±5°C Ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A9	Isolation	II	X	10	∞	1	∞	10	∞	1	∞	MΩ
B1-B6	Output short circuit stress	III	S	1	2	1	2	1	2	1	2	Minute
C1	Continuity	IV	S	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	Ω
C2-C5	Continuity	IV	X	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	Ω
D1,D2,D4, D5,D7,D8, D10,D11, D13,D14, D16,D17	DC gain accuracy	V	X	4.90	5.10	4.85	5.15	4.85	5.15	4.80	5.20	V
D3,D6,D9, D12,D15, D18	DC gain accuracy	V	X	4.95	5.05	4.92	5.08	4.92	5.08	4.90	5.10	V
E1-E6	AC gain accuracy	VI	X	-0.25	0.24	-0.38	0.36	-0.38	0.36	-0.50	0.48	dB
E7-E12	AC gain accuracy	VI	X	-0.26	0.26	-0.35	0.34	-0.35	0.34	-0.45	0.42	dB
E13-E18	Bandwidth	VI	X	75	∞	75	∞	75	∞	75	∞	kHz
F1-F6	Input offset voltage	VII	X	-5	5	-5	5	-6	6	-6	6	mV
G1-G6	Phase shift	VIII	X	← See table VIII →								
H1-H3	Delta phase shift	VIII	X	-0.5	0.5	-0.5	0.5	-0.5	0.5	-0.5	0.5	Degree
J1-J6	Output voltage swing	IX	X	6.8	10.6	6.7	10.6	6.7	10.6	6.6	10.6	Vrms
K1-K6	Output distortion	IX	X	0.0	2.0	0.0	2.0	0.0	2.0	0.0	2.0	Percent
L1-L12	Input resistance	X	S	19.8	20.2	19.7	20.3	19.7	20.3	19.6	20.4	kΩ
M1, M2	Power supply filter capacitance	XI	S	0.075	0.125	0.075	0.125	0.075	0.125	0.075	0.125	µF
N1, N2	Quiescent power supply current	XII	S	3.0	18.5	3.0	18.5	3.0	21.0	3.0	21.0	mA
P1-P6	Self noise	XIII	S	-∞	-96	-∞	-96	-∞	-96	-∞	-96	dBV
R1-R6	Power supply ripple rejection	XIV	Q	-∞	-30	-∞	-30	-∞	-30	-∞	-30	dBV

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° ±5°C Ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
S1-S6	Output impedance	XV	S	0	7.5	0	10	0	10	0	15	Ω
T1-T12	Output slew rate	XVI	S	0.3	∞	0.3	∞	0.3	∞	0.3	∞	V/µs
U1-U6	Crosstalk	XVII	Q	-∞	-46	-∞	-46	-∞	-46	-∞	-46	dB

NOTES:
1. (X) parameters shall be 100 percent tested during module acceptance tests.
2. (S) parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3. (Q) parameters shall be tested during qualification testing only.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life.

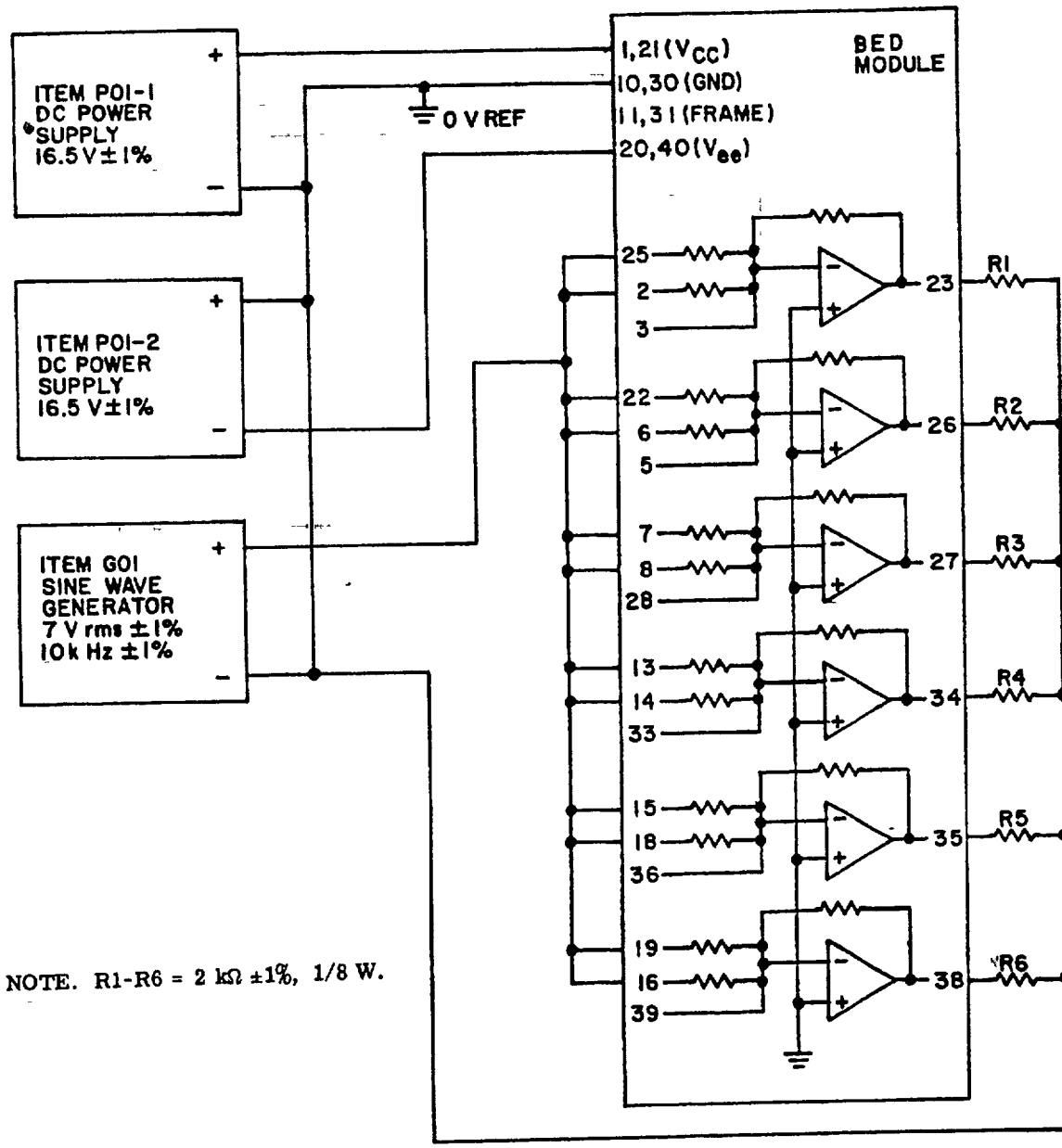
6.3 Cross reference of classification. Part number M28787/205-1 supersedes part number M28787/205. The two parts are interchangeable.

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5963-N032)

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NOTE. R1-R6 = 2 kΩ ±1%, 1/8 W.

FIGURE 2. Operating temperature and component temperature test circuit.

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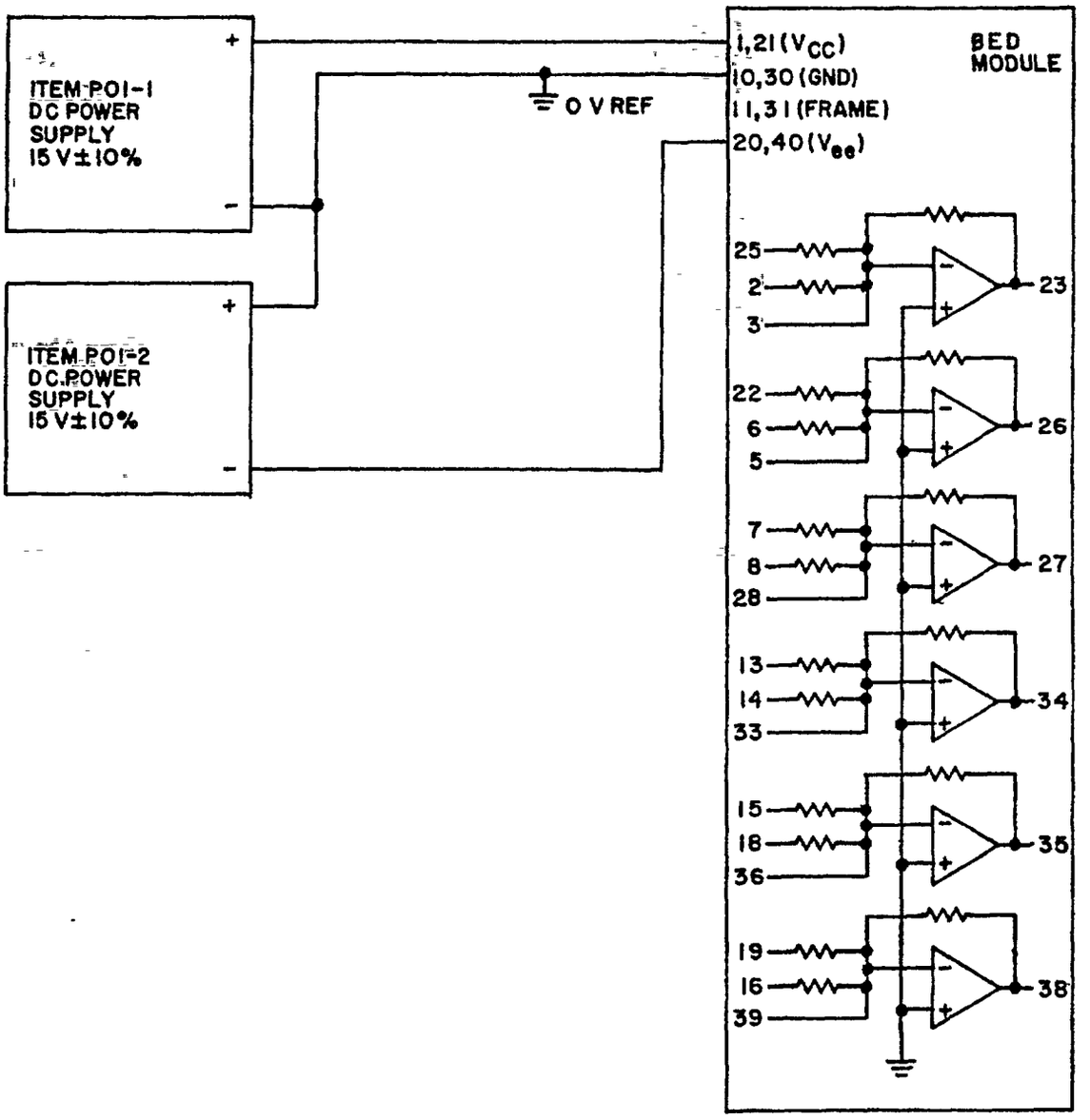
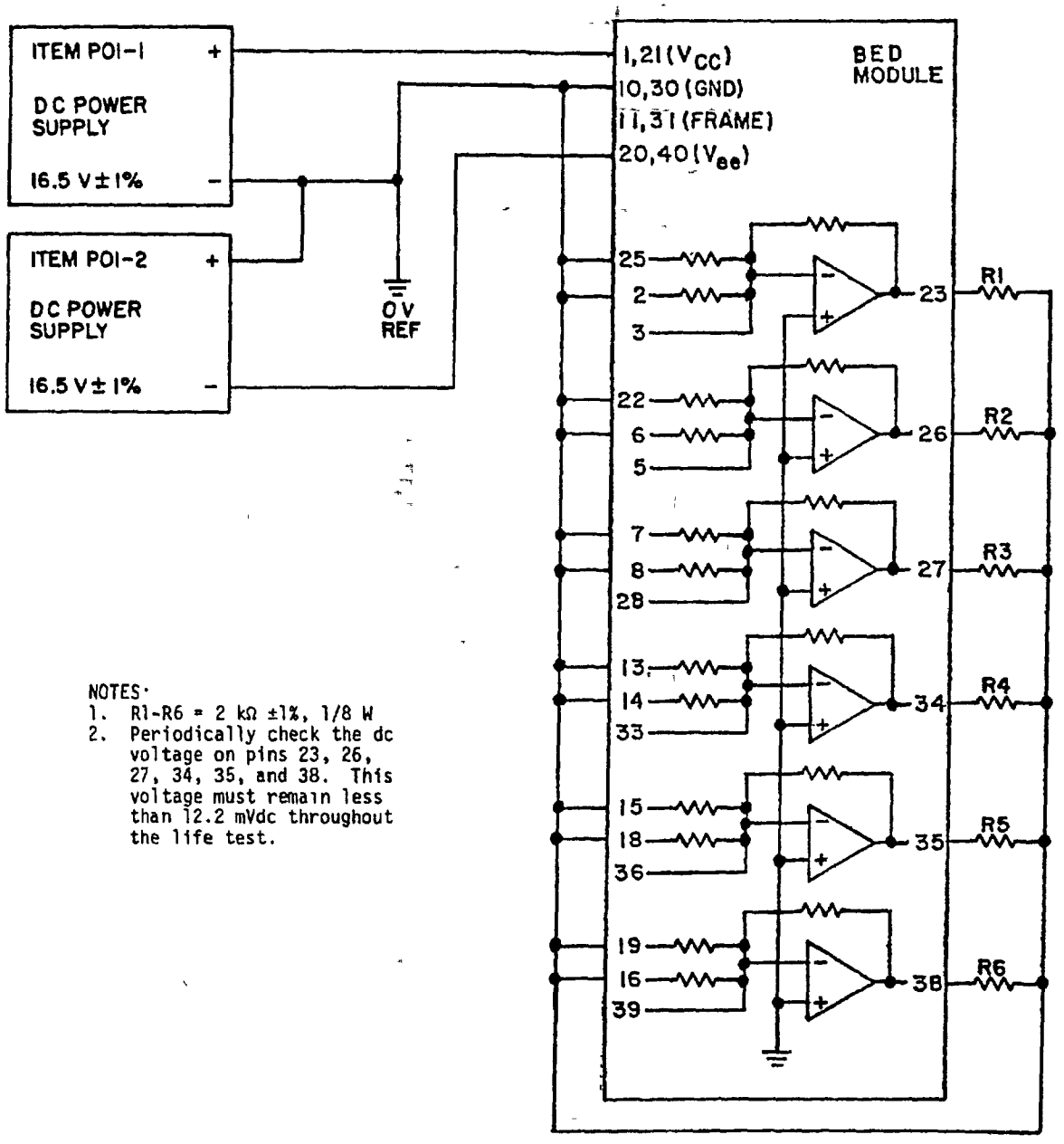


FIGURE 3. Vibration test circuit.

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NOTES:
1. R1-R6 = 2 kΩ ±1%, 1/8 W
2. Periodically check the dc voltage on pins 23, 26, 27, 34, 35, and 38. This voltage must remain less than 12.2 mVdc throughout the life test.

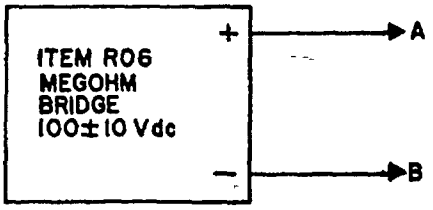
FIGURE 4. Life test circuit.

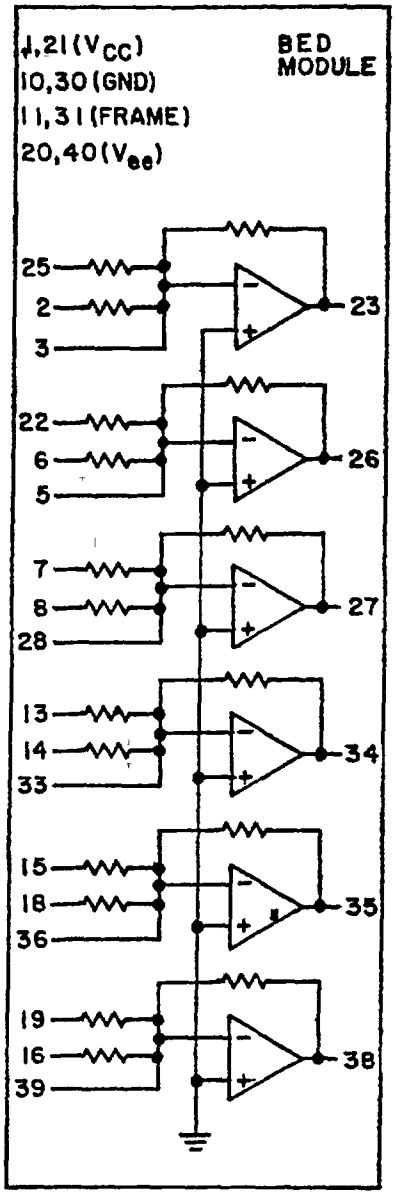
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TABLE II. Isolation test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation, frame	1-10,12-30,32-40	11,31	10	∞	MΩ	5
A2	Isolation, used pins to unused pins	1,21,10,30,20,40,11, 31,25,2,3,22,6,5,7, 8,28,13,14,33,15, 18,36,19,16,39,23, 26,27,34,35,38	4,9,12,17,24,29,32, 37	↓	↓	↓	↓
A3	Isolation, unused pins	4	9,12,17,24,29,32,37				
A4	↓	9	12,17,24,29,32,37				
A5		12	17,24,29,32,37				
A6		17	24,29,32,37				
A7		24	29,32,37				
A8		29	32,37				
A9	Isolation, unused pins	32	37	10	∞	MΩ	5

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

FIGURE 5. Isolation test circuit for table II.

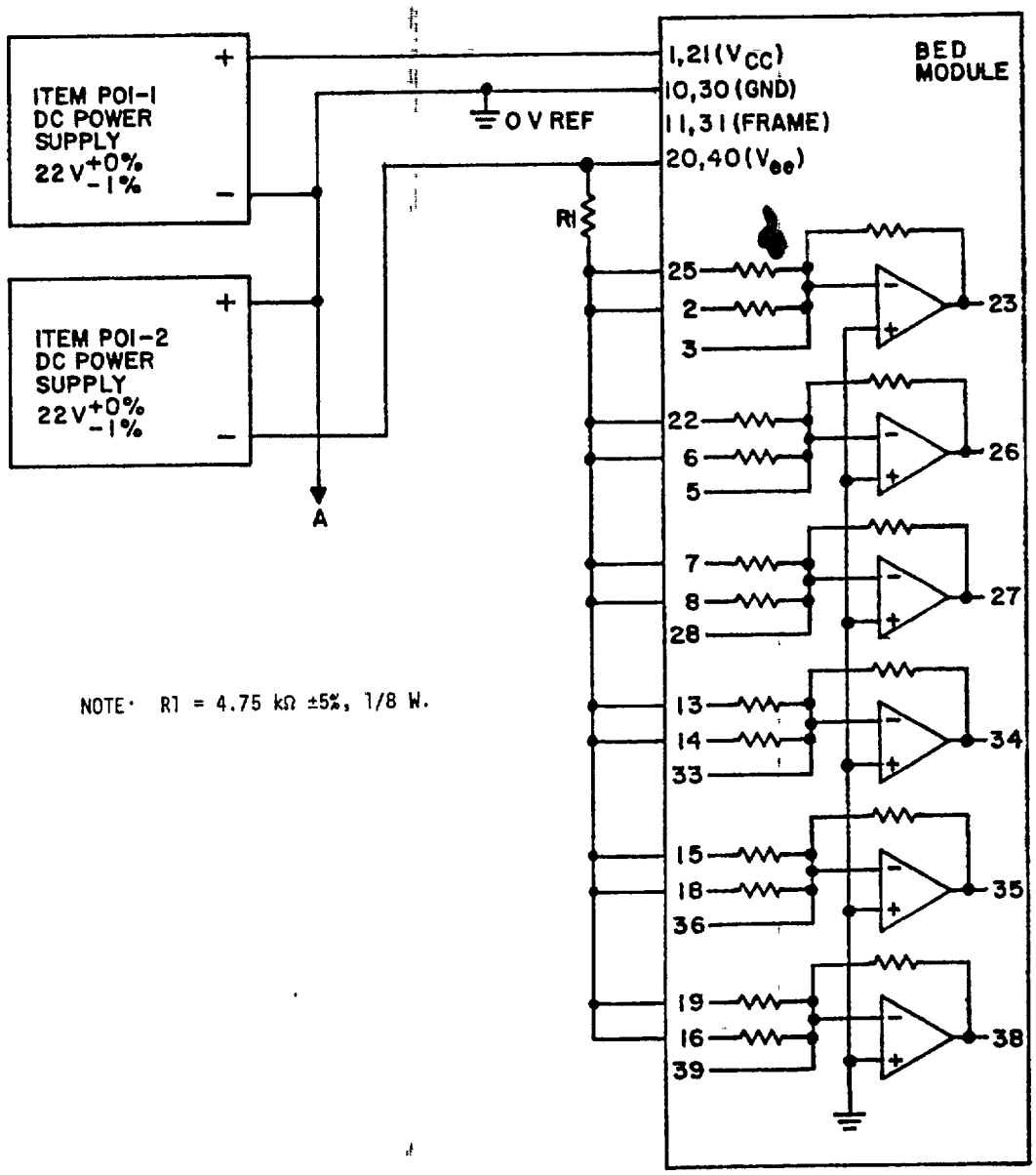
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TABLE III. Output short circuit stress test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
B1	Output short circuit stress (see note)	23	1	2	Minute	6
B2	↓	26	↓	↓	↓	↓
B3		27				
B4		34				
B5		35				
B6	Output short circuit stress (see note)	38	1	2	Minute	6

NOTE: The module meets the electrical requirements of test condition B if it passes test conditions C through U in the order specified in 4.4.2 following subjection to test condition B.

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NOTE: R1 = 4.75 kΩ ±5%, 1/8 W.

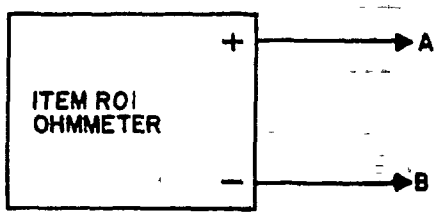
FIGURE 6. Output short circuit stress test circuit for table III.

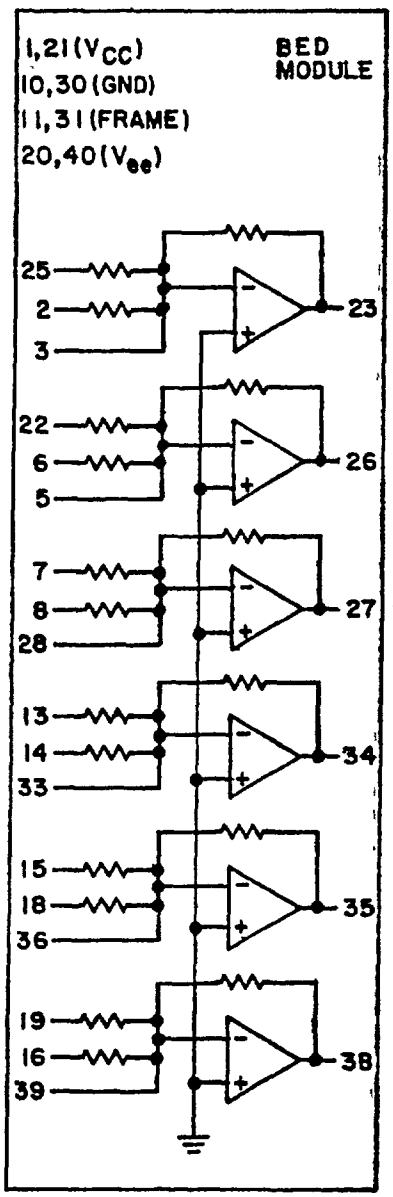
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TABLE IV. Continuity test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	Continuity	11,31	Frame	0.0	1.0	Ω	7
C2	↓	11	31	↓	↓	↓	↓
C3	↓	1	21	↓	↓	↓	↓
C4	↓	10	30	↓	↓	↓	↓
C5	Continuity	20	40	0.0	1.0	Ω	7

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NOTE: On test condition C1 (when measuring to frame), penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

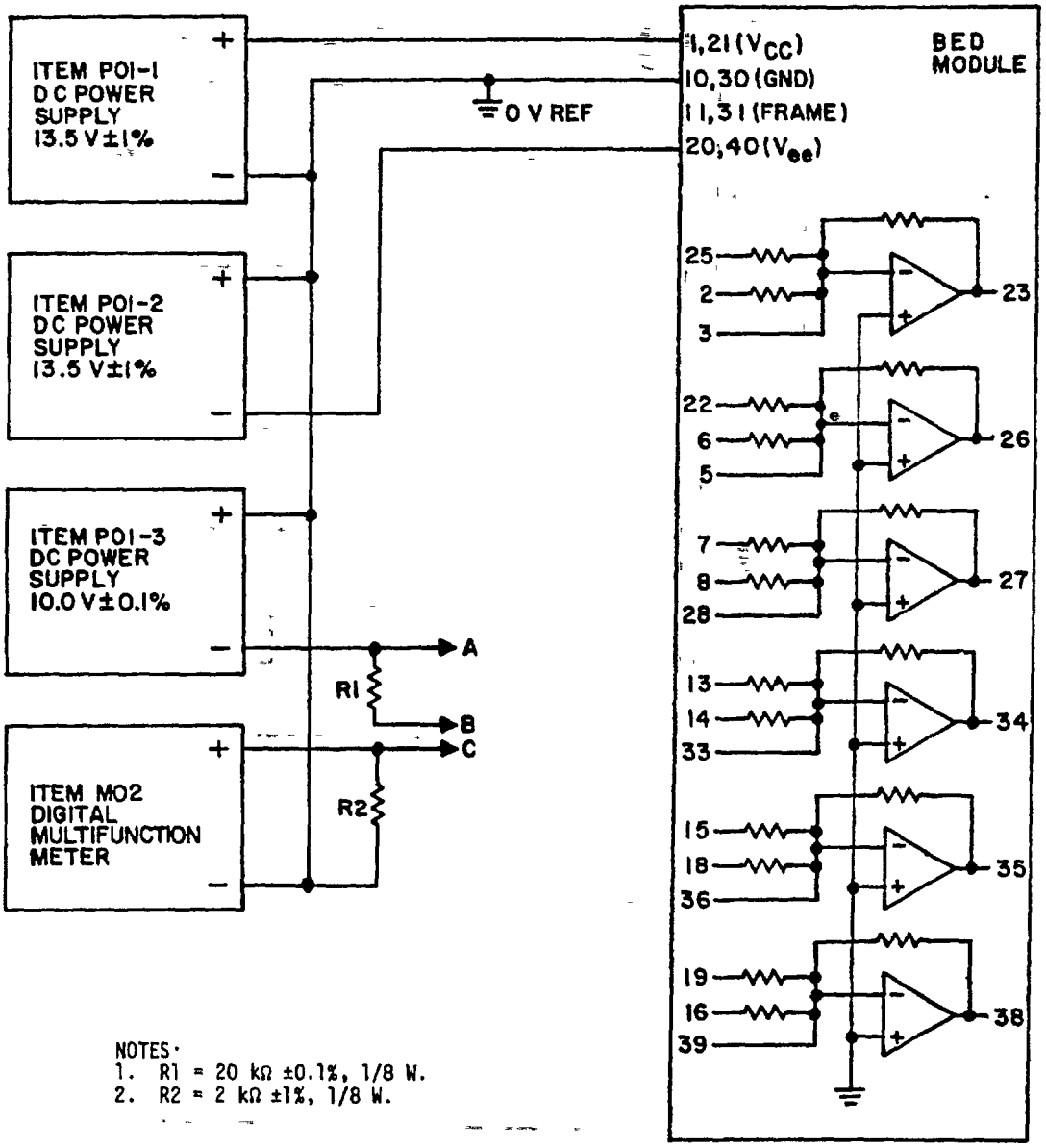
FIGURE 7. Continuity test circuit for table IV.

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TABLE V. DC gain accuracy test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
D1	DC gain accuracy	25	Open	23	4.90	5.10	V	8
D2	↓	2	Open	23	4.90	5.10	↓	↓
D3		Open	3	23	4.95	5.05		
D4		22	Open	26	4.90	5.10		
D5		6	Open	26	4.90	5.10		
D6		Open	5	26	4.95	5.05		
D7		7	Open	27	4.90	5.10		
D8		8	Open	27	4.90	5.10		
D9		Open	28	27	4.95	5.05		
D10		13	Open	34	4.90	5.10		
D11		14	Open	34	4.90	5.10		
D12		Open	33	34	4.95	5.05		
D13		15	Open	35	4.90	5.10		
D14		18	Open	35	4.90	5.10		
D15		Open	36	35	4.95	5.05		
D16		19	Open	38	4.90	5.10		
D17		16	Open	38	4.90	5.10		
D18	DC gain accuracy	Open	39	38	4.95	5.05	V	8

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NOTES:
1. R1 = 20 kΩ ±0.1%, 1/8 W.
2. R2 = 2 kΩ ±1%, 1/8 W.

FIGURE 8. DC gain accuracy test circuit for table V.

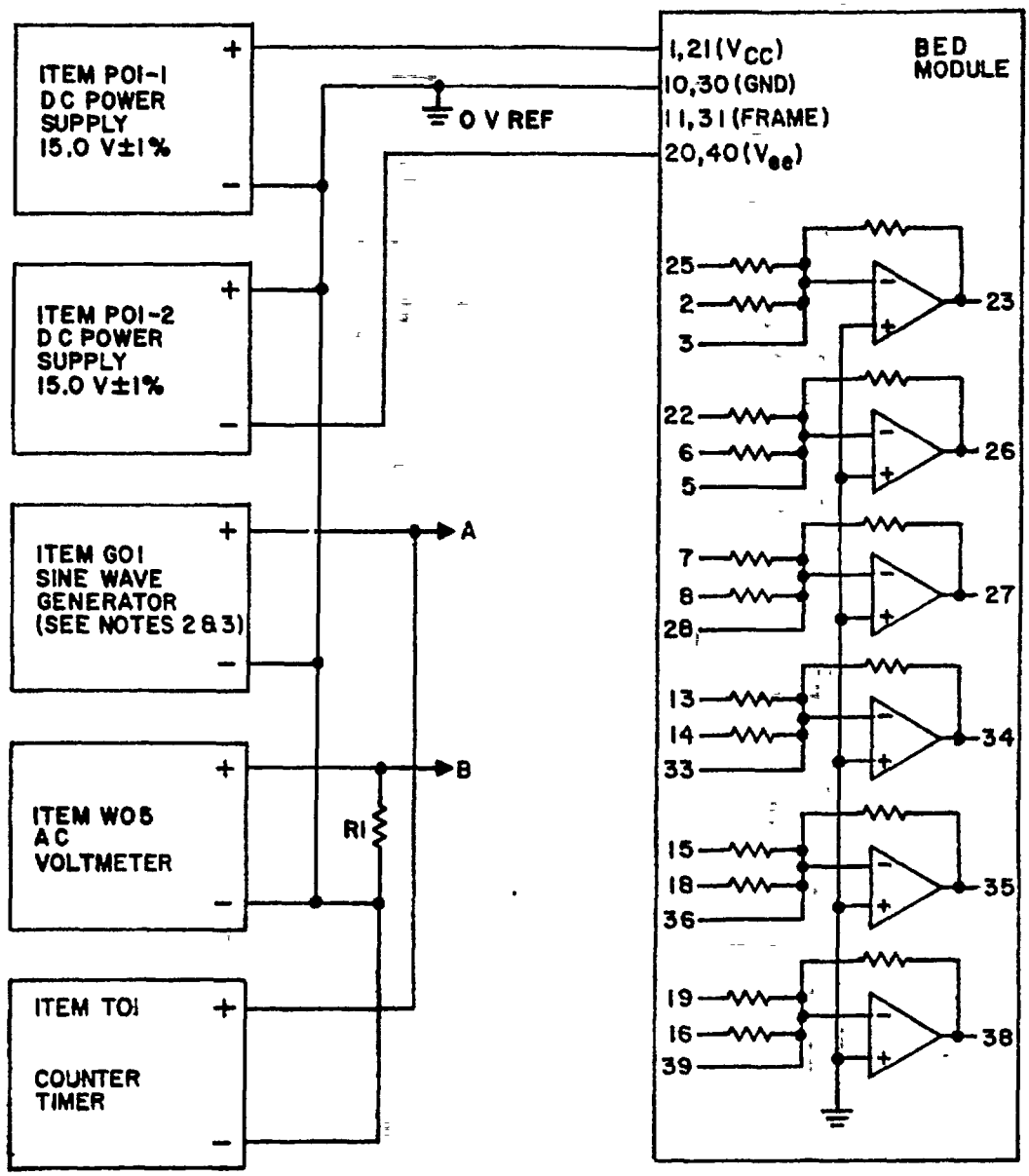
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TABLE VI. AC gain accuracy and bandwidth test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
E1	AC gain accuracy (see note)	2,25	23	6.8	7.2	Vrms	9
E2		6,22	26				
E3		7,8	27				
E4		13,14	34				
E5		15,18	35				
E6		16,19	38	6.8	7.2	Vrms	
E7		2,25	23	97	103	mVrms	
E8		6,22	26				
E9		7,8	27				
E10		13,14	34				
E11		15,18	35				
E12	AC gain accuracy (see note)	16,19	38	97	103	mVrms	
E13	Bandwidth	2,25	23	75	∞	kHz	
E14		6,22	26				
E15		7,8	27				
E16		13,14	34				
E17		15,18	35				
E18	Bandwidth	16,19	38	75	∞	kHz	9

NOTE: $20 \log \frac{\text{(Test condition E)}}{\text{Vrms (Item G01)}} = \text{dB}$

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NOTES:
1. R1 = 2 kΩ ±1%, 1/8 W.
2. Item G01 setup:
   a. 7 Vrms ±1% for test conditions E1-E6.
   b. 100 mVrms ±1% for test conditions E7-E18.
   c. Set frequency to 1 kHz ±1% for test conditions E1-E6. Set frequency to 10 kHz ±1% for test conditions E7-E12.
   d. For each test condition E13-E18, increase the frequency of item G01 until the voltage on item W05 equals 0.708 times the reading for test conditions E7-E12, respectively.
3. Whenever the frequency of item G01 is changed the amplitude must be rechecked.

FIGURE 9. AC gain accuracy and bandwidth test circuit for table VI.

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TABLE VII. Output offset voltage test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
F1	Output offset voltage (see note)	23	-10.1	10.1	mV	10
F2	↓	26	↓	↓	↓	↓
F3		27				
F4		34				
F5		35				
F6	Output offset voltage (see note)	38	-10.1	10.1	mV	10

NOTE: Output offset voltage $= \frac{R_f + R_I}{R_I}$ (input offset voltage)

$= 2.02$ (input offset voltage)

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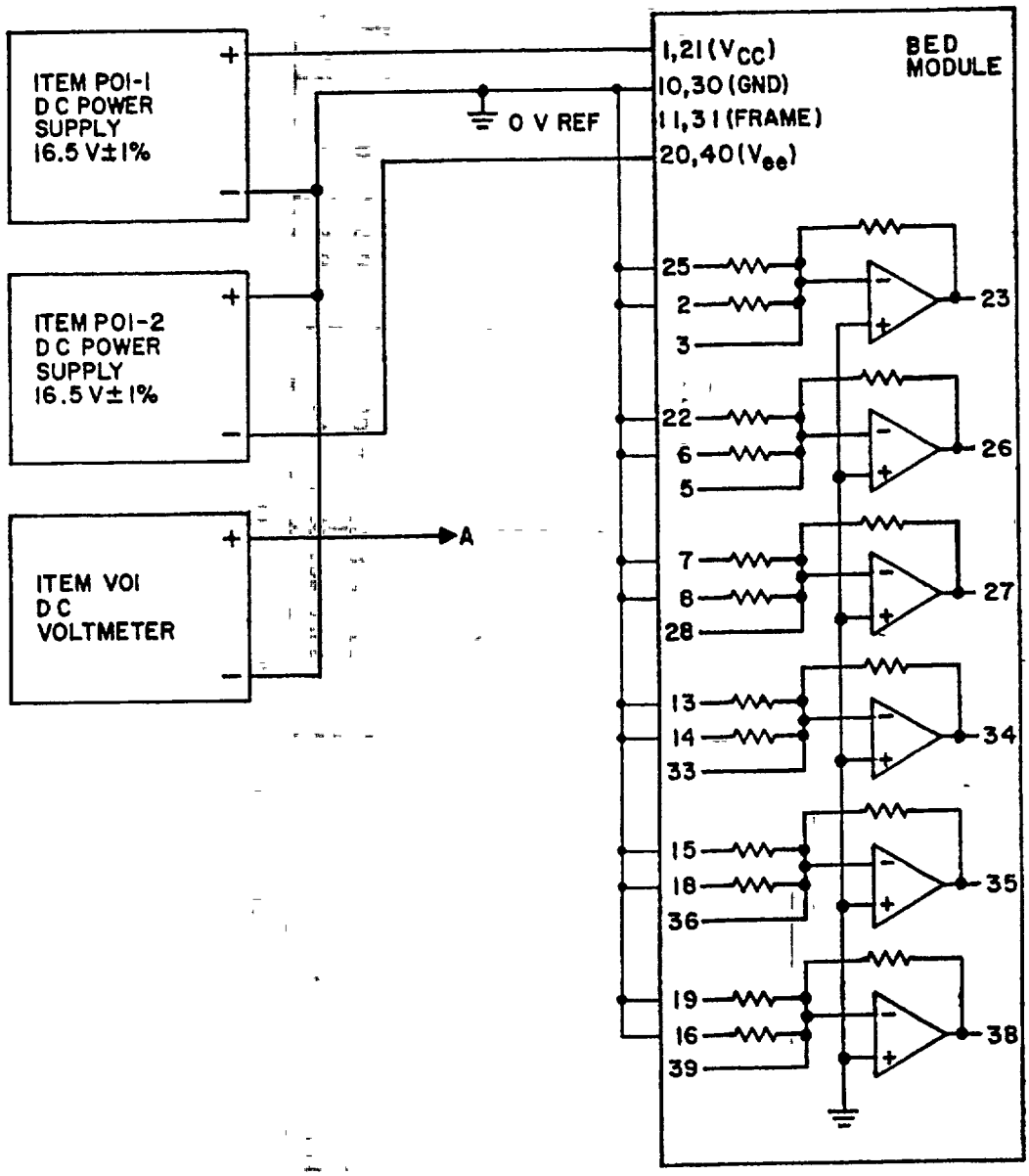


FIGURE 10. Output offset voltage test circuit for table VII.

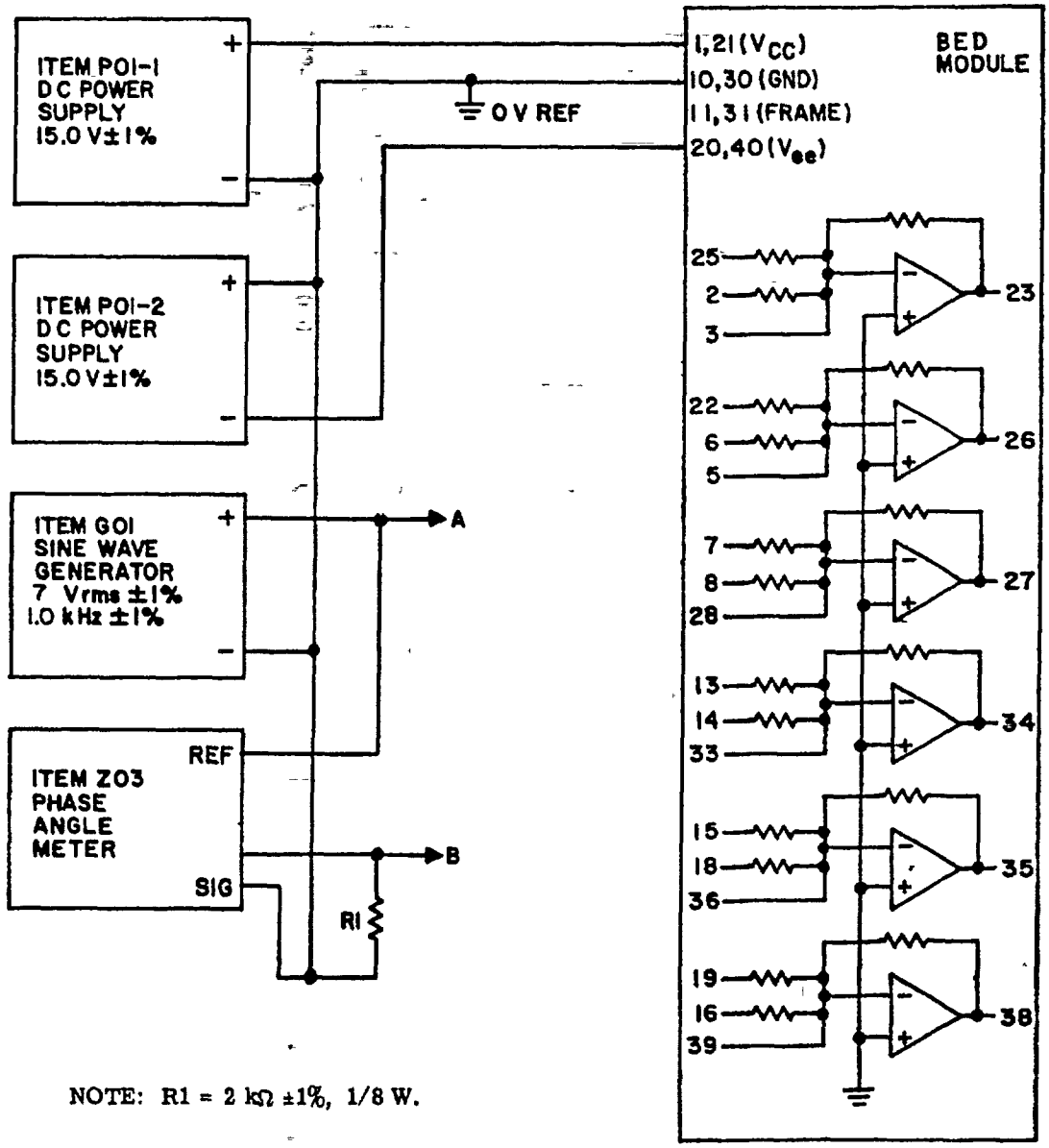
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TABLE VIII. Phase shift and delta phase shift test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
G1	Phase shift	2, 25	23	See note	See note	Degree	11
G2	↓	6, 22	26	↓	↓	↓	↓
G3	↓	7, 8	27	↓	↓	↓	↓
G4	↓	13, 14	34	↓	↓	↓	↓
G5	↓	15, 18	35	↓	↓	↓	↓
G6	Phase shift	16, 19	38	See note	See note	↓	↓
H1	Delta phase shift	Subtract G1 from G2		-0.5	0.5	↓	↓
H2	Delta phase shift	Subtract G3 from G4		-0.5	0.5	↓	↓
H3	Delta phase shift	Subtract G5 from G6		-0.5	0.5	Degree	11

NOTE: Record the no fail data for the test condition H calculations.

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NOTE: R1 = 2 kΩ ±1%, 1/8 W.

FIGURE 11. Phase shift and delta phase shift test circuit for table VIII.

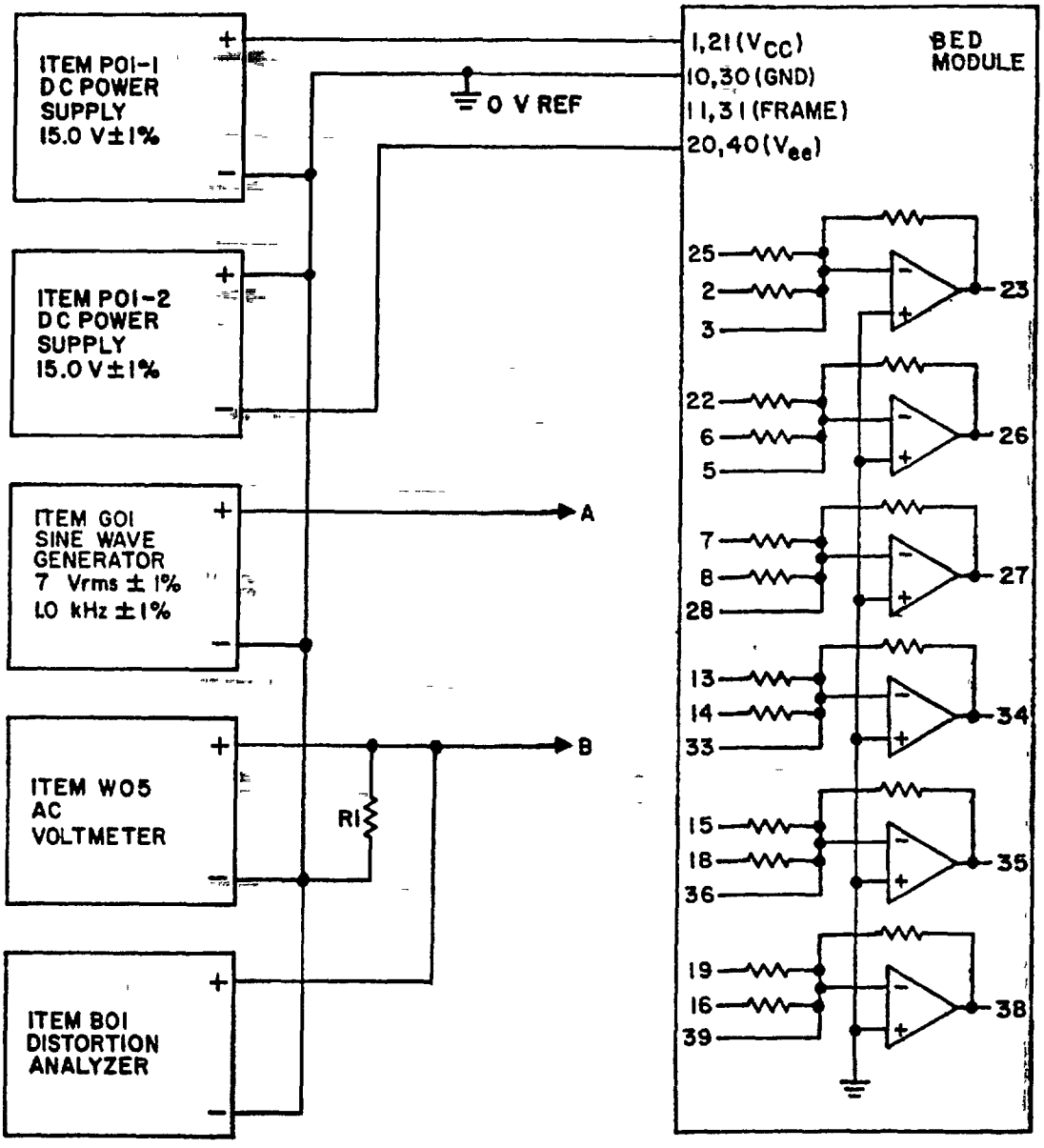
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TABLE IX. Output voltage swing and output distortion test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J1	Output voltage swing	2, 25	23	6.8	10.6	Vrms	12
J2	↓	6, 22	26	↓	↓	↓	↓
J3		7, 8	27				
J4		13, 14	34				
J5		15, 18	35				
J6	Output voltage swing	16, 19	38	6.8	10.6	Vrms	
K1	Output distortion (D) (see note)	2, 25	23	0.0	2.0	Percent	
K2	↓	6, 22	26	↓	↓	↓	
K3		7, 8	27				
K4		13, 14	34				
K5		15, 18	35				
K6	Output distortion (D) (see note)	16, 19	38	0.0	2.0	Percent	12

NOTE: Output distortion (D) = $D_o - D_i$ where D_o is the distortion reading on the output and D_i is the distortion reading on the input.

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NOTE: R1 = 2 kΩ ±1%, 1/8 W.

FIGURE 12. Output voltage swing and output distortion test circuit for table IX.

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TABLE X. Input resistance test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
L1	Input resistance	25	3	19.8	20.2	kΩ	13
L2	↓	2	3	↓	↓	↓	↓
L3		22	5				
L4		6	5				
L5		7	28				
L6		8	28				
L7		13	33				
L8		14	33				
L9		15	36				
L10		18	36				
L11		19	39				
L12	Input resistance	16	39	19.8	20.2	kΩ	13

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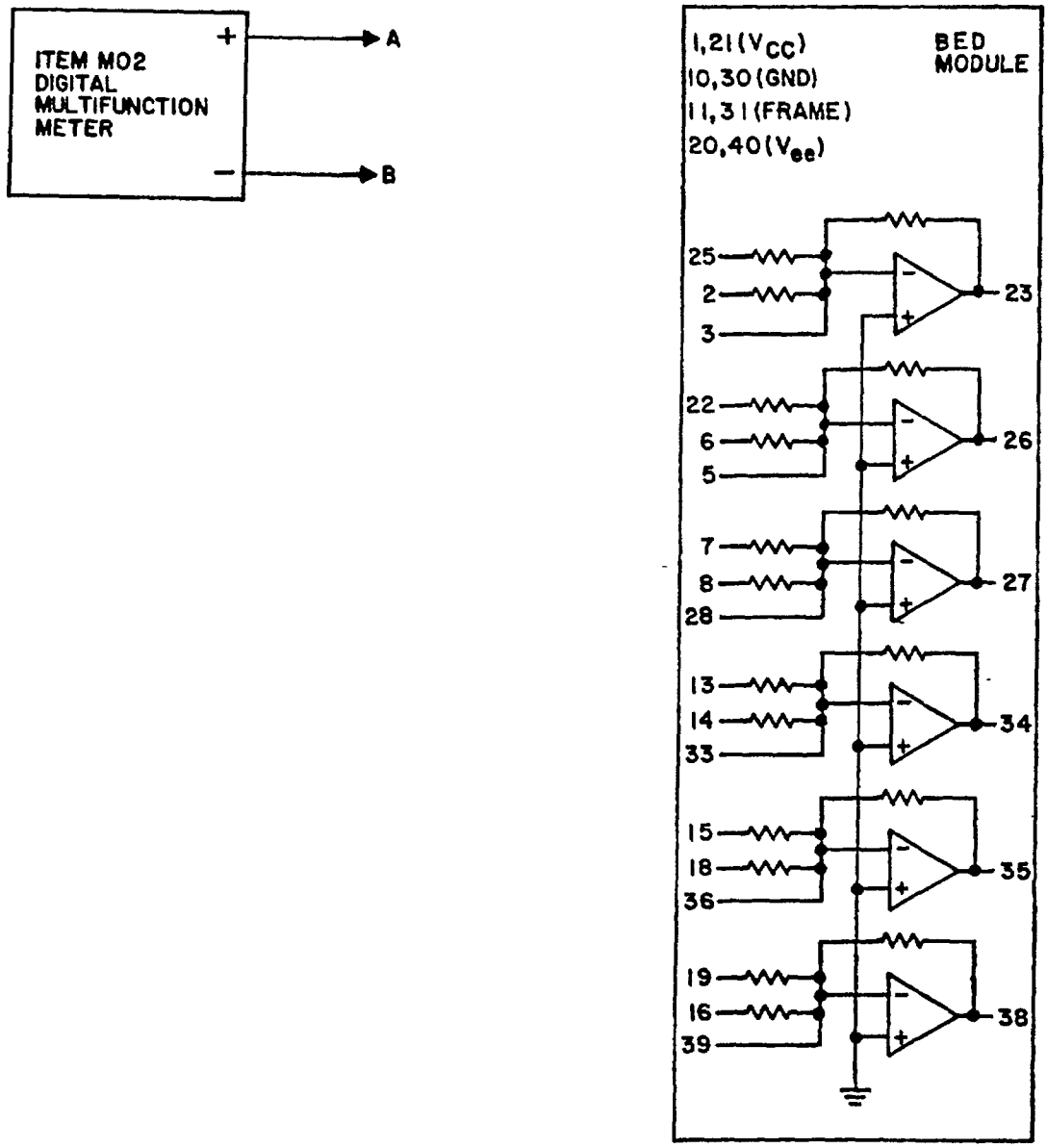


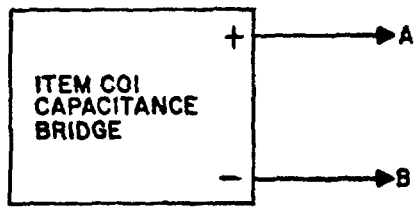
FIGURE 13. Input resistance test circuit for table X.

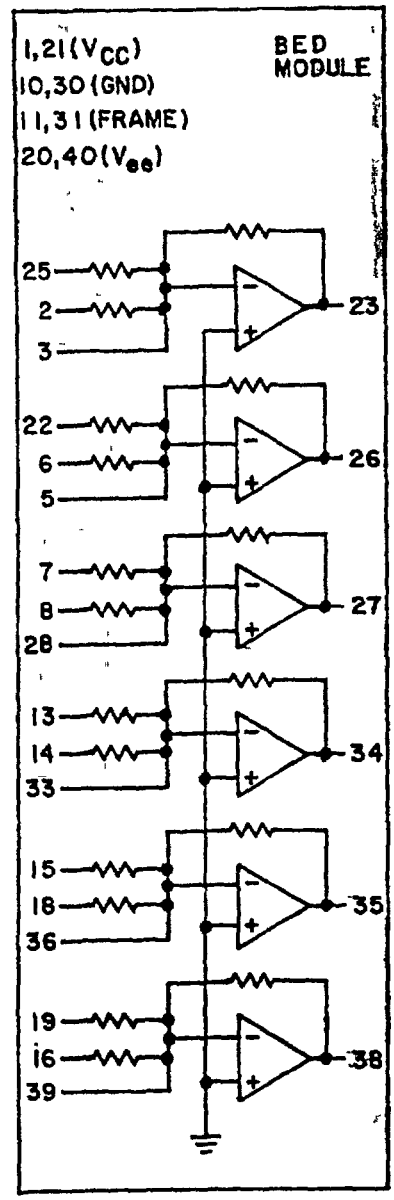
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TABLE XI. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
M1	Power supply filter capacitance	1, 21	10, 30	0.075	0.125	µF	14
M2	Power supply filter capacitance	10, 30	20, 40	0.075	0.125	µF	14

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NOTE: Item C01 setup:
V_{pk} = 0.3 V
Frequency = 120 Hz.

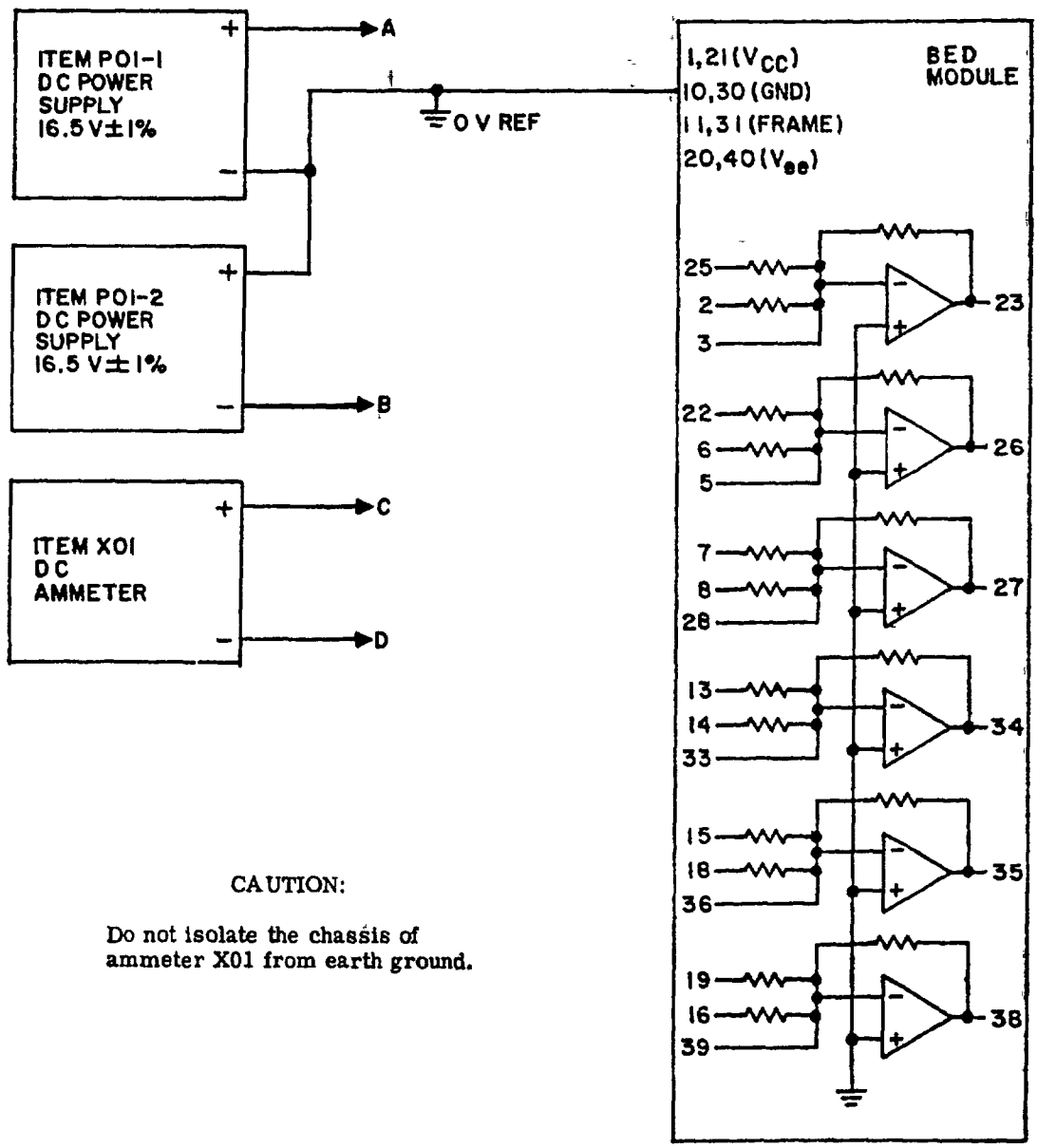
FIGURE 14. Power supply filter capacitance test circuit for table XI.

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TABLE XII. Quiescent power supply current test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
N1	Quiescent power supply current	C	20,40	A	1,21	3.0	18.5	mA	15
N2	Quiescent power supply current	1,21	D	20,40	B	3.0	18.5	mA	15

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

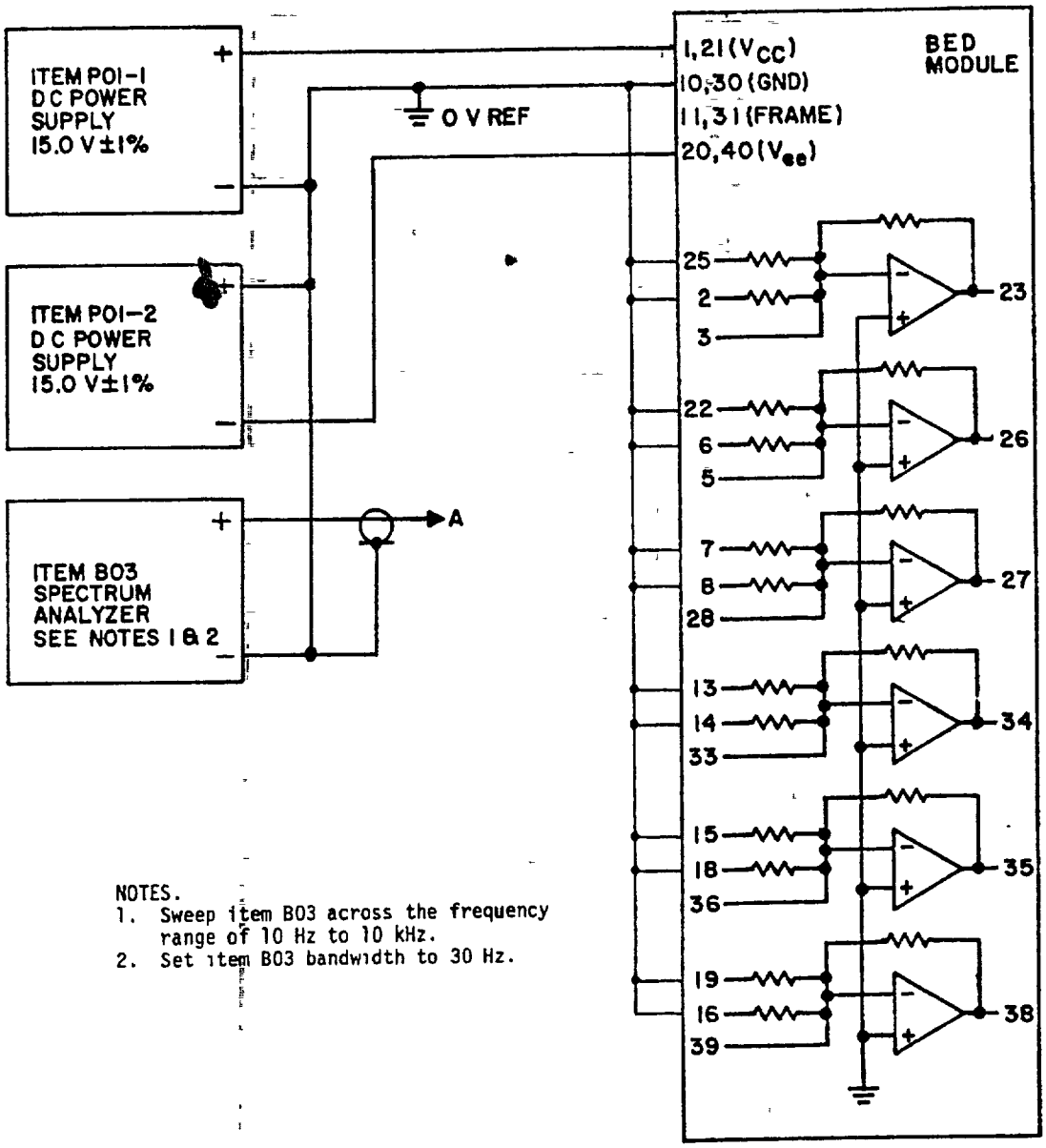
FIGURE 15. Quiescent power supply current test circuit for table XII.

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TABLE XIII. Self noise test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
P1	Self noise	23	-∞	-96	dBV	16
P2	↓	26	↓	↓	↓	↓
P3		27				
P4		34				
P5		35				
P6	Self noise	38	-∞	-96	dBV	16

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NOTES.
1. Sweep item B03 across the frequency range of 10 Hz to 10 kHz.
2. Set item B03 bandwidth to 30 Hz.

FIGURE 16. Self noise test circuit for table XIII.

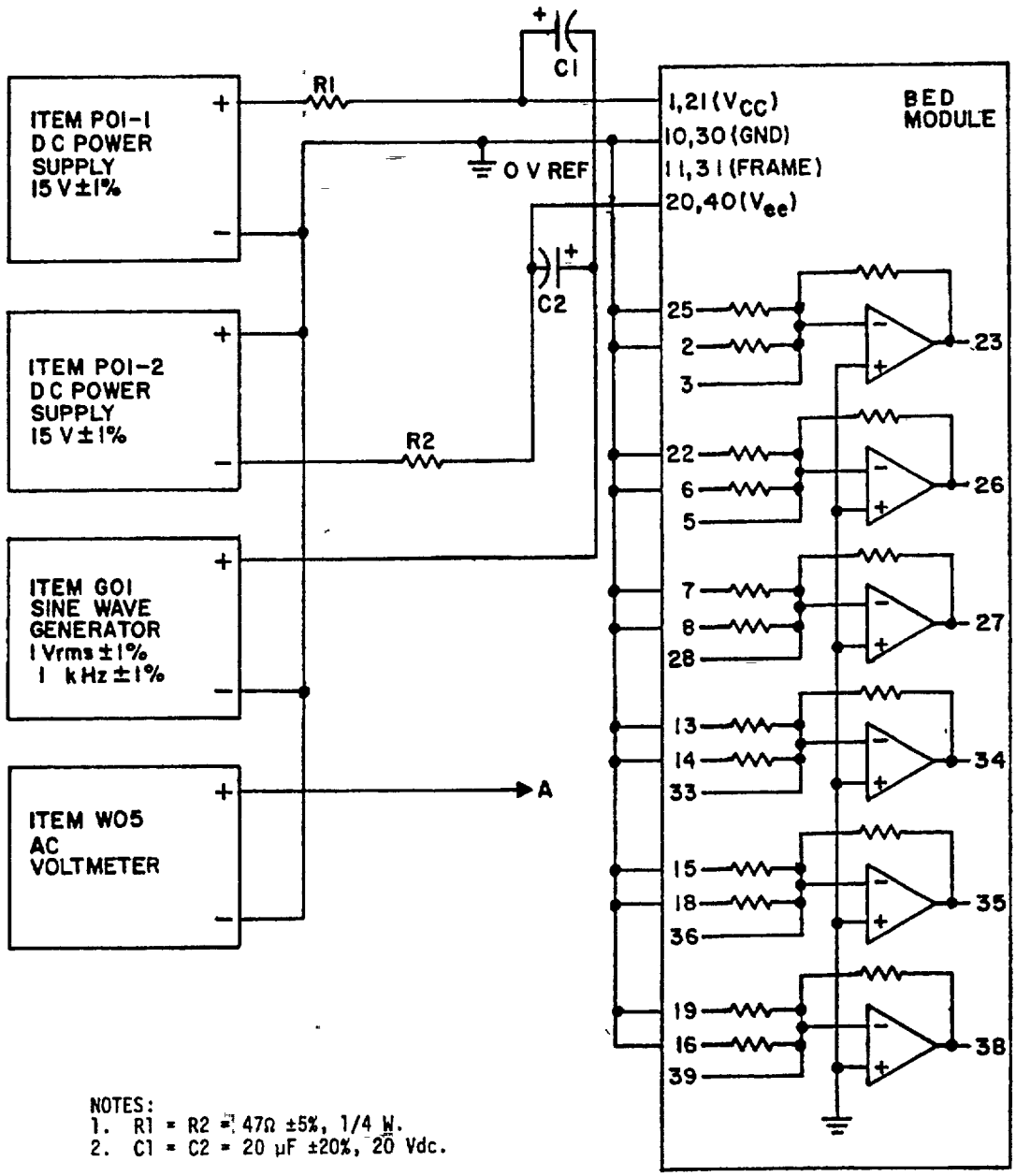
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TABLE XIV. Power supply ripple rejection test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
R1	Power supply ripple rejection (see note)	23	0	31.6	mVrms	17
R2	↓	26	↓	↓	↓	↓
R3		27				
R4		34				
R5		35				
R6	Power supply ripple rejection (see note)	38	0	31.6	mVrms	17

NOTE: $20 \log \frac{\text{(Test condition R)}}{1 \text{ Vrms}} = \text{dBV}$

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NOTES:
1. R1 = R2 = 47Ω ±5%, 1/4 W.
2. C1 = C2 = 20 µF ±20%, 20 Vdc.

FIGURE 17 Power supply ripple rejection test circuit for table XIV.

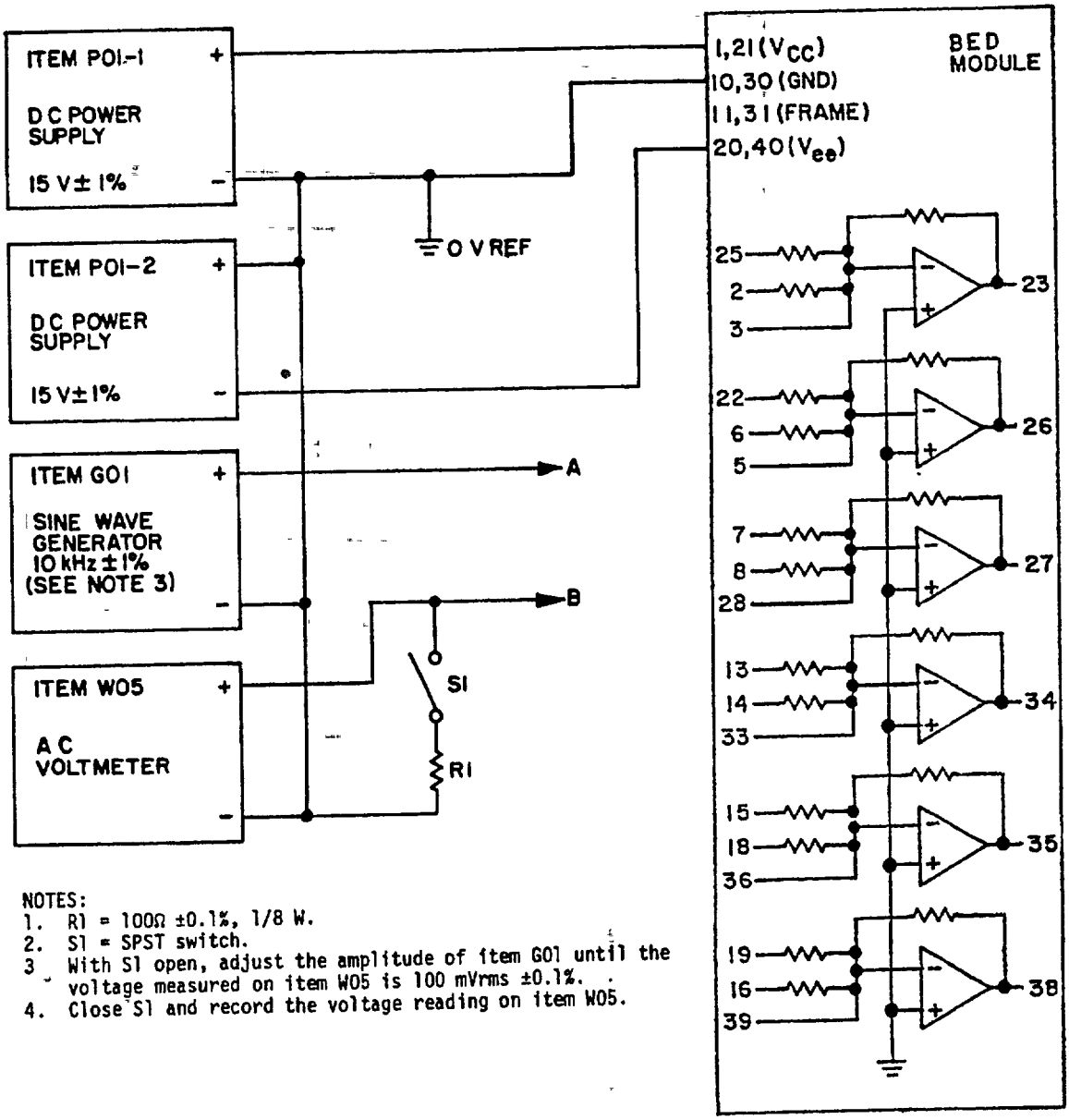
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TABLE XV. Output impedance test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
S1	Output impedance (see note)	2, 25	23	93	100	mVrms	18
S2	↓	6, 22	26	↓	↓	↓	↓
S3	↓	7, 8	27	↓	↓	↓	↓
S4	↓	13, 14	34	↓	↓	↓	↓
S5	↓	15, 18	35	↓	↓	↓	↓
S6	Output impedance (see note)	16, 19	38	93	100	mVrms	18

NOTE: $\frac{100\ \text{mVrms}}{100\Omega + \text{Output impedance}} = \frac{\text{(Test condition S)}}{100\Omega}$

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NOTES:
1. R1 = 100Ω ±0.1%, 1/8 W.
2. S1 = SPST switch.
3. With S1 open, adjust the amplitude of item GO1 until the voltage measured on item WO5 is 100 mVrms ±0.1%.
4. Close S1 and record the voltage reading on item WO5.

FIGURE 18. Output impedance test circuit for table XV.

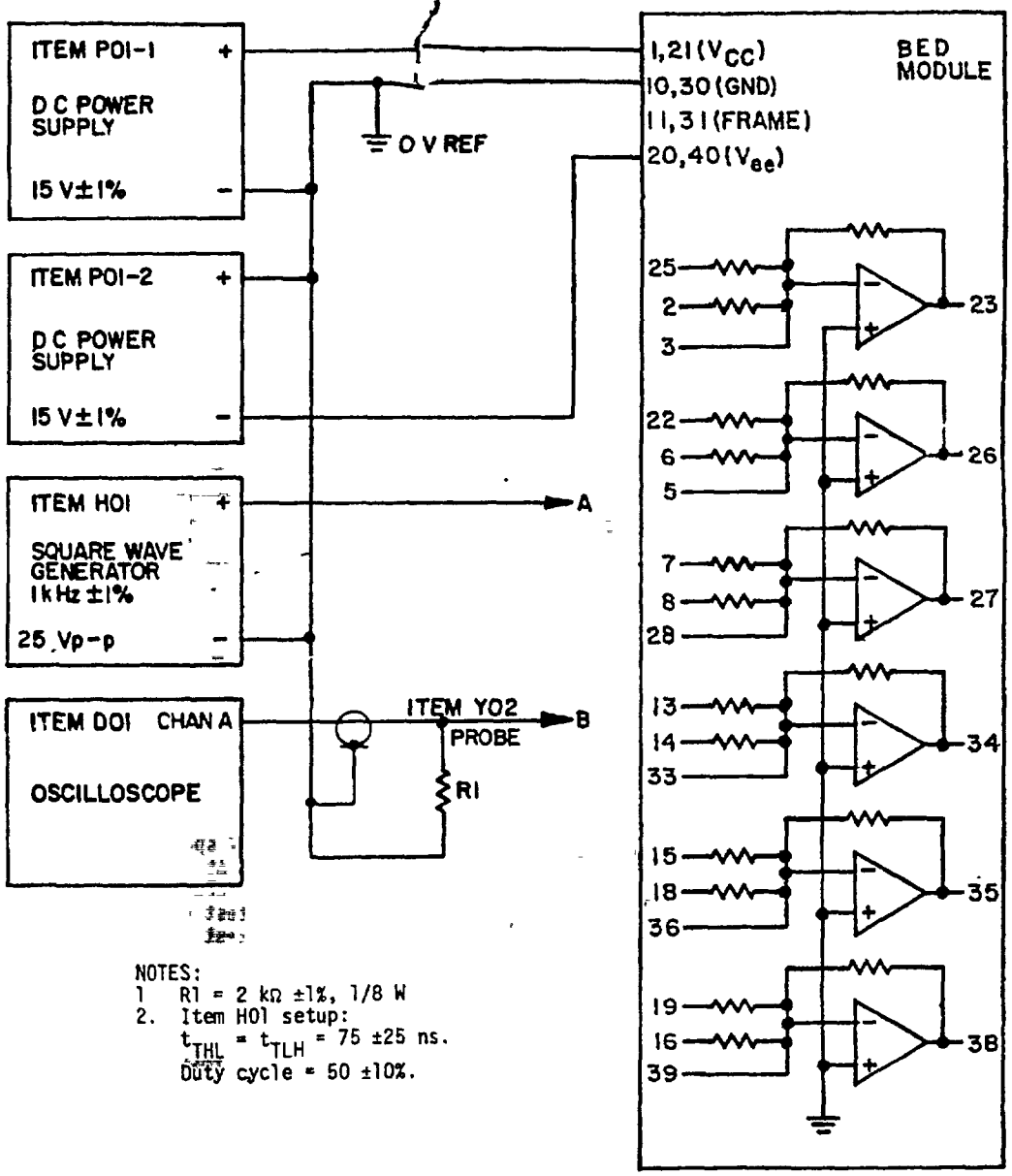
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TABLE XVI. Output slew rate test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
T1	Output slew rate (see note)	2, 25	23	0.0	53.4	µs	19, 20
T2	↓	2, 25	23	↓	↓	↓	↓
T3		6, 22	26				
T4		6, 22	26				
T5		7, 8	27				
T6		7, 8	27				
T7		13, 14	34				
T8		13, 14	34				
T9		15, 18	35				
T10		15, 18	35				
T11		16, 19	38				
T12	Output slew rate (see note)	16, 19	38	0.0	53.4	µs	19, 20

NOTE: $\frac{0.3\ V}{1\ \mu s} = \frac{2\ V}{\text{(Test condition T maximum)}}$

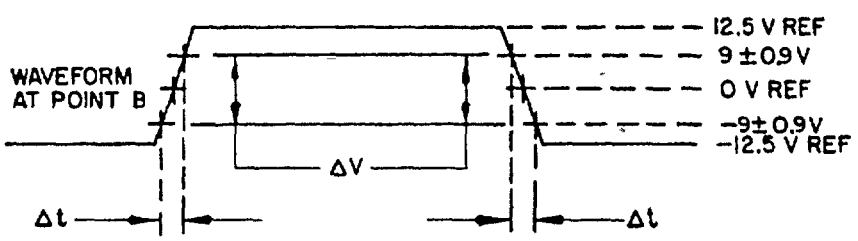
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NOTES:
1. R1 = 2 kΩ ±1%, 1/8 W
2. Item HO1 setup:
   $t_{THL} = t_{TLH} = 75 \pm 25$ ns.
   Duty cycle = 50 ±10%.

FIGURE 19. Output slew rate test circuit for table XVI.

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NOTE: The output slew rate is ΔV/Δt. With the ΔV specified as 18 V ±1.8 V, the only parameter that needs to be measured is the Δt. The Δts for both the positive and negative transitions will be recorded for each output. The Δts for the positive transitions will be for test conditions T1, T3, T5, T7, and T9 and T11 and the negative transitions will be for test conditions T2, T4, T6, T8, T10, and T12.

FIGURE 20. Output slew rate test circuit waveform for table XVI.

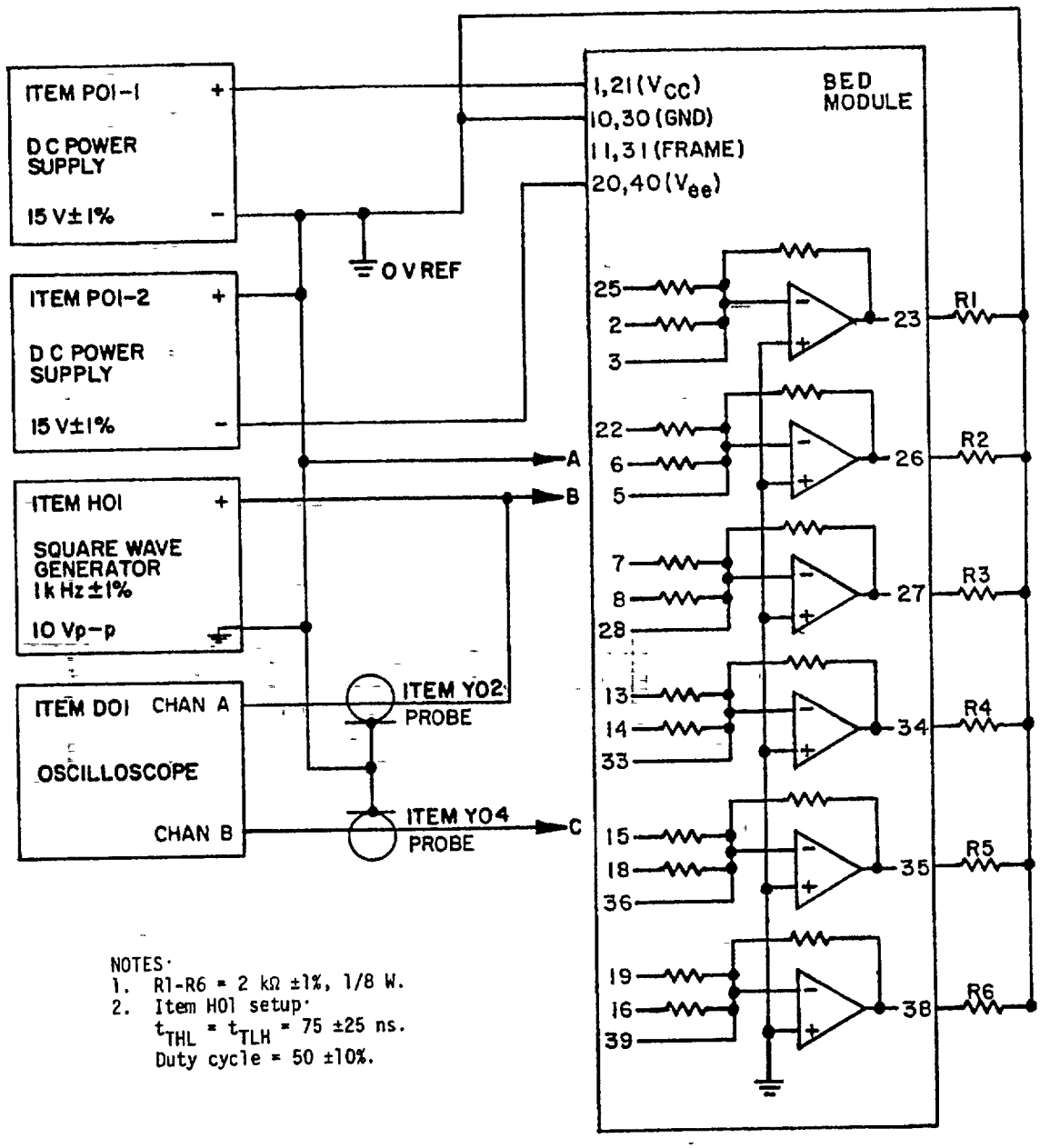
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TABLE XVII. Crosstalk test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
U1	Crosstalk (see note)	2, 25	6,22,7,8, 13,14,15, 18,16,19	23	0	50	mV	21, 22
U2		6, 22	2,25,7,8, 13,14,15, 18,16,19	26				
U3		7, 8	2,25,6,22, 13,14,15, 18,16,19	27				
U4		13, 14	2,25,6,22, 7,8,15,18, 16,19	34				
U5		15, 18	2,25,6,22, 7,8,13,14, 16, 19	35				
U6	Crosstalk (see note)	16, 19	2,25,6,22, 7,8,13,14, 15,18	38	0	50	mV	21, 22

NOTE: $20 \log \frac{\text{(Test condition U)}}{10\ \text{V}} = \text{dB}$

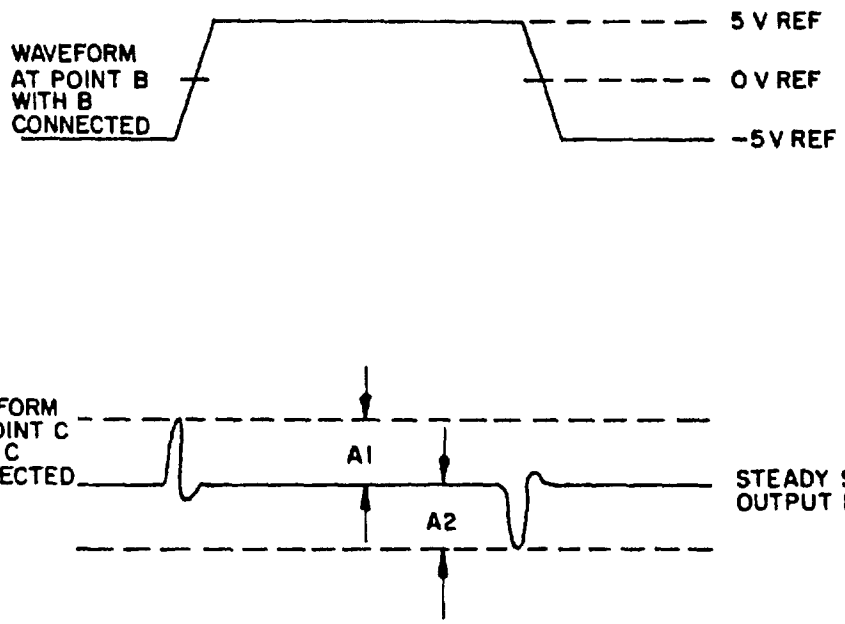
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NOTES:
1. R1-R6 = 2 kΩ ±1%, 1/8 W.
2. Item H01 setup:
   $t_{THL} = t_{TLH} = 75 \pm 25$ ns.
   Duty cycle = 50 ±10%.

FIGURE 21. Crosstalk test circuit for table XVII.

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NOTE. The recorded data for test condition U will be the largest of the two readings A1 and A2.

FIGURE 22. Crosstalk test circuit waveforms for table XVII.

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